



Sweet Cherry Rutin- and Horse Chestnut B-escin-Functionalized Gold and Silver Nanoparticles for Broad-Spectrum Inhibition of SARS-CoV-2

Ilyes Zatl^{1*}

¹ PhD in Microbiology, Independent Researcher, Tlemcen, 13000, Algeria.

E-mail: ilyes.zatl@aol.com

Tel: +213-54031-5422

ORCID: 0000-0003-0476-8765

Address: Cité Vingt Villas n°14, Oudjida, Tlemcen, Algeria



1. INTRODUCTION & AIM

SARS-CoV-2 variants continue to pose global health threats due to mutations, higher transmissibility and immune escape. Natural biomolecules and metallic nanoparticles offer promising complementary antiviral strategies. This study aims to evaluate, via *in silico* approaches, the potential of rutin (from sweet cherry) and B-escin (from horse chestnut), alone and in combination with gold (AuNPs) and silver (AgNPs) nanoparticles, against key targets of SARS-CoV-2 and its major variants.



2. ABSTRACT

The continual emergence of SARS-CoV-2 variants (Alpha, Beta, Gamma, Delta, Omicron) demands broad-spectrum antiviral strategies able to overcome mutations, increased transmissibility, and immune escape. Here we present an *in silico* nanobiotechnology framework that evaluates natural biomolecules and metallic nanoparticles against multiple SARS-CoV-2 spike variants. We assessed the antiviral potential of B-escin (a triterpenoid saponin), rutin (a flavonoid), gold nanoparticles (AuNPs), and silver nanoparticles (AgNPs) using molecular docking, molecular dynamics simulations, and quantum mechanical analyses implemented with AutoDock 4.2 and Schrodinger server. Targets included the spike glycoprotein receptor-binding domain (RBD) complexed with human ACE2, the main protease (M^{pro}), and RNA-dependent RNA polymerase (RdRp). Rutin and B-escin both showed consistently favorable, variant-dependent binding to spike-ACE2 interfaces, forming multiple hydrogen bonds and stabilizing interactions; rutin demonstrated notable structural adaptability across variant-specific binding pockets. B-escin exhibited broad-spectrum inhibitory signatures with differential susceptibility among variants, while AgNPs and AuNPs displayed complementary target preferences—AgNPs favored RdRp interactions and AuNPs bound more strongly to spike and ACE2-spike complexes. The Omicron spike presented the highest ACE2 binding affinity in our models. Combining B-escin and rutin with metallic nanoparticles enhanced predicted antiviral potency, target specificity, and molecular stability relative to single agents, with nanoparticle physicochemical properties (size, morphology, surface functionalization) strongly influencing interaction profiles and synergistic effects. Based on these computational results, we propose a biomolecule-functionalized metallic nanoparticle platform for localized nasal prophylaxis and therapy against respiratory coronaviruses. Experimental and clinical validation will be required to confirm translational potential.

Keywords: SARS-CoV-2; COVID-19; Variants; Rutin; B-escin; Gold nanoparticles; Silver nanoparticles; Molecular docking; Molecular dynamics; Nanobiotechnology; Nasal delivery

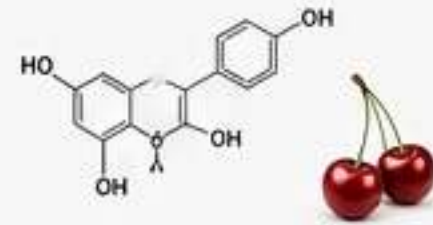
3. BIOCOMPONENTS & NANOPARTICLES

Natural biomolecules

RUTIN

Flavonoid (sweet cherry)

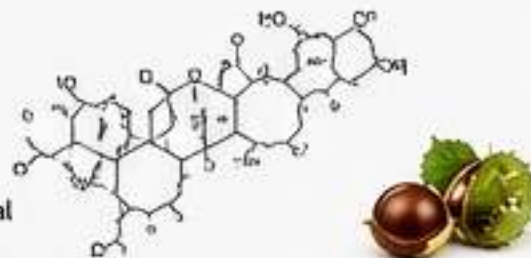
- Antioxidant, anti-inflammatory and antiviral activities



B-ESCIN

Triterpenoid saponin (horse chestnut)

- Membrane-active, anti-inflammatory and antiviral properties



Metallic nanoparticles



AuNPs

- High stability
- Strong affinity for spike/ACE2 targets
- Biocompatible



AgNPs

- Potent antiviral activity
- Favors RdRp binding
- Broad-spectrum potential

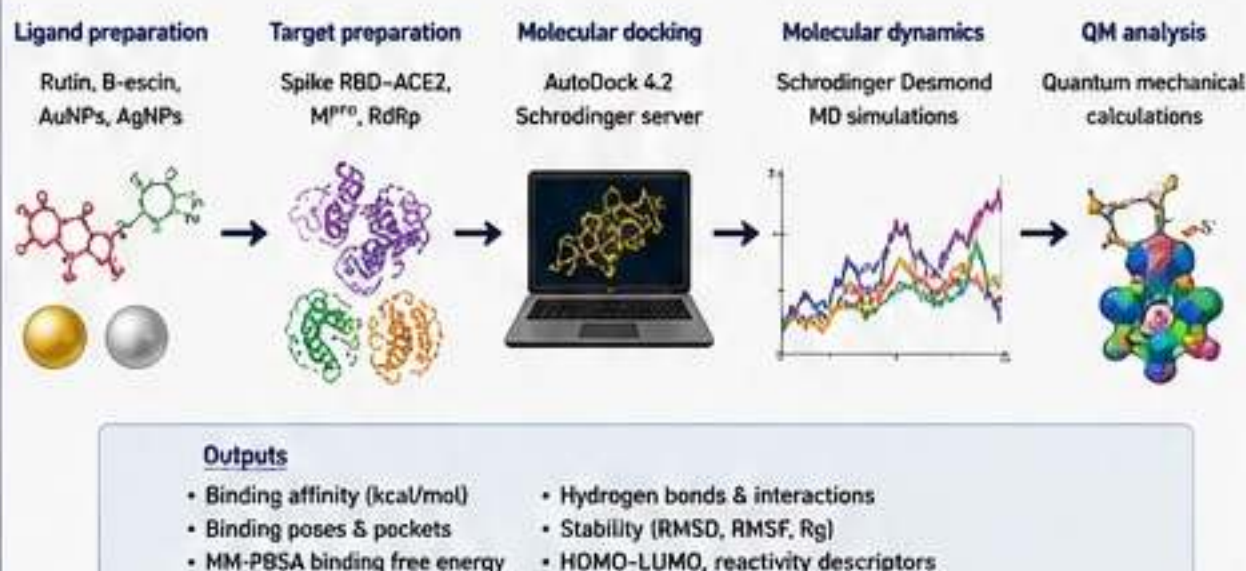
4. TARGETS

Spike RBD – ACE2 interface
Critical for viral entry into host cells

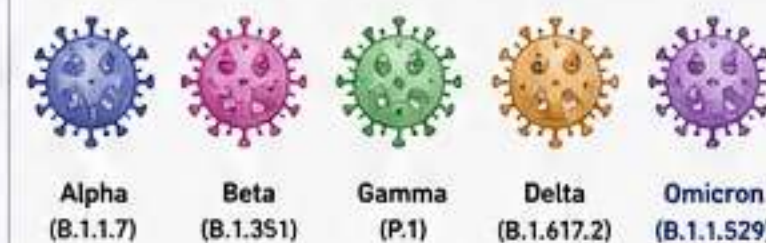
Main protease (M^{pro})
Essential for viral polyprotein cleavage and replication

RNA-dependent RNA polymerase (RdRp)
Key enzyme for viral genome replication and transcription

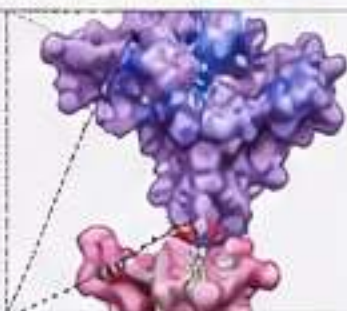
5. IN SILICO METHODOLOGY



6. SARS-CoV-2 VARIANTS STUDIED



Omicron spike exhibited the highest ACE2 binding affinity in our models.



7. MOLECULAR DOCKING RESULTS (ΔG_{bind} , kcal/mol)

Spike RBD-ACE2 Complex (Average Binding Affinity)

Ligand	Alpha	Beta	Gamma	Delta	Omicron
Rutin	-9.45	-9.88	-9.70	-9.51	-10.62
B-escin	-9.82	-10.21	-9.96	-9.77	-10.85
AuNPs	-10.35	-10.79	-10.61	-10.32	-11.48
AgNPs	-9.21	-9.64	-9.38	-9.11	-9.92

Omicron spike shows the highest affinity with ACE2 and with most tested ligands.

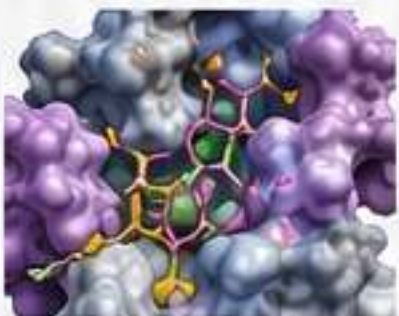
Target Preference of Nanoparticles (Average ΔG_{bind})

Target	AgNPs (favored)	AuNPs (favored)
Spike RBD-ACE2	-9.48	-10.67
M ^{pro}	-8.35	-9.38
RdRp	-10.28	-8.96

AgNPs preferentially interact with RdRp, while AuNPs show stronger binding to spike and ACE2-spike complexes.

8. BINDING INTERACTIONS (Examples)

Rutin – Omicron Spike RBD-ACE2

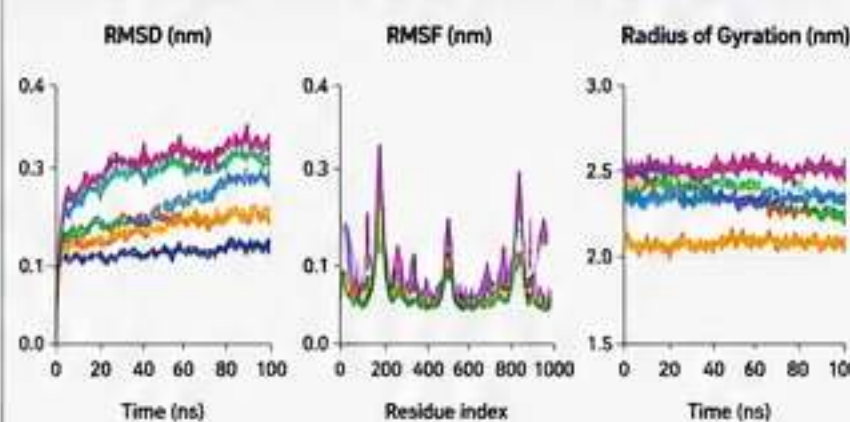


B-escin – Delta Spike RBD-ACE2



Both rutin and B-escin establish multiple stabilizing interactions within variant-specific pockets.

9. MOLECULAR DYNAMICS & STABILITY



Complexes with rutin/B-escin and nanoparticles show higher stability (lower RMSD/RMSF) and compact structures (lower Rg) during 100 ns simulations.

10. PROPOSED APPLICATION

Nasal Prophylaxis & Therapy Platform



- Localized delivery to respiratory mucosa
- Reduced systemic exposure
- Enhanced retention with nanoparticles
- Broad-spectrum inhibition of variants
- Synergistic action of biomolecules + NPs



11. KEY FINDINGS

- Rutin and B-escin bind favorably to spike-ACE2 interfaces across variants, forming multiple stabilizing interactions.
- Rutin shows high structural adaptability to variant-specific binding pockets.
- B-escin exhibits broad-spectrum antiviral signatures with variable susceptibility among variants.
- AgNPs preferentially target RdRp, whereas AuNPs bind more strongly to spike and ACE2-spike complexes.
- Combining biomolecules with AuNPs/AgNPs enhances antiviral potency, target specificity, and molecular stability.
- Nanoparticle properties (size, morphology, surface functionalization) strongly influence interaction profiles and synergistic effects.
- Proposed biomolecule-functionalized nanoparticle platform for nasal delivery as a promising strategy against SARS-CoV-2 and other respiratory coronaviruses.

12. FUTURE PERSPECTIVES

- Experimental validation (in vitro antiviral assays)
- In vivo safety, pharmacokinetics and efficacy studies
- Formulation optimization for nasal delivery
- Evaluation against emerging coronaviruses
- Clinical development of biocompatible nano-formulations

REFERENCES (Selected)

- Zatl I, Boublenza L. Battling COVID-19 leveraging nanobiotechnology: Gold and silver nanoparticle – B-escin conjugates as SARS-CoV-2 inhibitors. Open Life Sci. 2025;20(1):20221047.
- Ilyes Zatl et al. Computational and nanobiotechnological approaches against SARS-CoV-2. (Manuscript in preparation).
- Relevant literature on rutin, B-escin, AuNPs/AgNPs and SARS-CoV-2 targets.